

1) PROGRAMACIÓN LINEAL GRÁFICA

Resuelva los siguientes problemas para X_1 y X_2 no negativas.

1.1)

$$\begin{array}{rcl} X_1 & & \leq 3 \\ & X_2 & \leq 6 \\ 6 X_1 + 4 X_2 & & \leq 36 \\ Z = 8 X_1 + 3 X_2 & & \text{MAX} \end{array}$$

1.2)

$$\begin{array}{rcl} -2 X_1 + X_2 & & \leq 2 \\ X_1 - X_2 & & \leq 2 \\ X_1 + X_2 & & \leq 5 \\ Z = 5 X_1 + 2 X_2 & & \text{MAX} \end{array}$$

1.3)

$$\begin{array}{rcl} & X_2 & \leq 3 \\ 4 X_1 + 6 X_2 & & \leq 24 \\ 4 X_1 - 3 X_2 & & \leq 12 \\ Z = 5 X_1 + 2 X_2 & & \text{MAX} \end{array}$$

1.4)

$$\begin{array}{rcl} 6 X_1 + 5 X_2 & & \leq 30 \\ & X_2 & \geq 1 \\ -2 X_1 + 2 X_2 & & \leq 6 \\ Z = 5 X_1 + 8 X_2 & & \text{MAX} \end{array}$$

✓ 1.5) hecho en simplex
con programa

$$\begin{array}{rcll}
 X_1 + X_2 & \leq & 300 \\
 2,5 X_1 + 4 X_2 & \leq & 1000 \\
 X_2 & = & 200 \\
 X_1 & \leq & 200
 \end{array}
 \left\{ \begin{array}{l} X_1 + X_2 + X_3 = 300 \\ 2,5 X_1 + 4 X_2 + X_4 = 1000 \\ X_2 = 200 \\ X_1 + X_5 = 200 \end{array} \right.$$

$$Z = 6 X_1 + 2 X_2 \quad \text{MAX}$$

1.6)
con programa

$$\begin{array}{rcll}
 X_2 & \leq & 3 \\
 4 X_1 + 6 X_2 & \leq & 24 \\
 2 X_1 + 2 X_2 & \geq & 0
 \end{array}$$

$$Z = -2 X_1 + 4 X_2 \quad \text{MAX}$$

1.7)
con programa

$$\begin{array}{rcll}
 X_1 & \leq & 6 \\
 X_1 + X_2 & \leq & 8 \\
 X_1 + 2 X_2 & \leq & 12
 \end{array}$$

$$Z = 4 X_1 + 4 X_2 \quad \text{MAX}$$

1.8)
con programa

$$\begin{array}{rcll}
 2 X_1 + 4 X_2 & \leq & 48 \\
 4 X_1 + 2 X_2 & \leq & 60 \\
 3 X_1 & \leq & 45
 \end{array}$$

$$Z = 6 X_1 + 4 X_2 \quad \text{MAX}$$

1.9)
con programa

$$\begin{array}{rcll}
 -5 X_1 + 3 X_2 & \geq & 5 \\
 X_1 + X_2 & \leq & 4 \\
 2 X_1 + X_2 & \geq & 10
 \end{array}$$

$$Z = 2 X_1 + X_2 \quad \text{MAX}$$

" However, the problem is
infeasible "

1.10)

the problem is unbounded

$$\begin{array}{rcl} & X2 & \geq 2 \\ 4 X1 + 6 X2 & \geq 24 \\ 10 X1 - 30 X2 & \geq 30 \end{array}$$

$$Z = X1 + 8 X2 \quad \text{MAX}$$

1.11)

optimal solution is achieve

$$\begin{array}{rcl} X1 & \geq 2 \\ 2 X1 + X2 & \leq 10 \\ X1 + 2 X2 & \leq 8 \\ X2 & \geq 1 \end{array}$$

$$Z = X1 - 2 X2 \quad \text{MIN}$$

$$-X1 + 2X2 \quad \text{Max}$$

1.12)

problem infeasible

$$\begin{array}{rcl} X1 + X2 & \leq 6 \\ 2 X1 + 3 X2 & \leq 1 \\ - X1 + 2 X2 & \geq 8 \end{array}$$

$$Z = 3 X1 + X2 \quad \text{MAX}$$

1.13)

con max

$$\begin{array}{rcl} 2 X1 + X2 & \leq 8 \\ X1 + X2 & \leq 6 \\ X1 & \leq 4 \end{array}$$

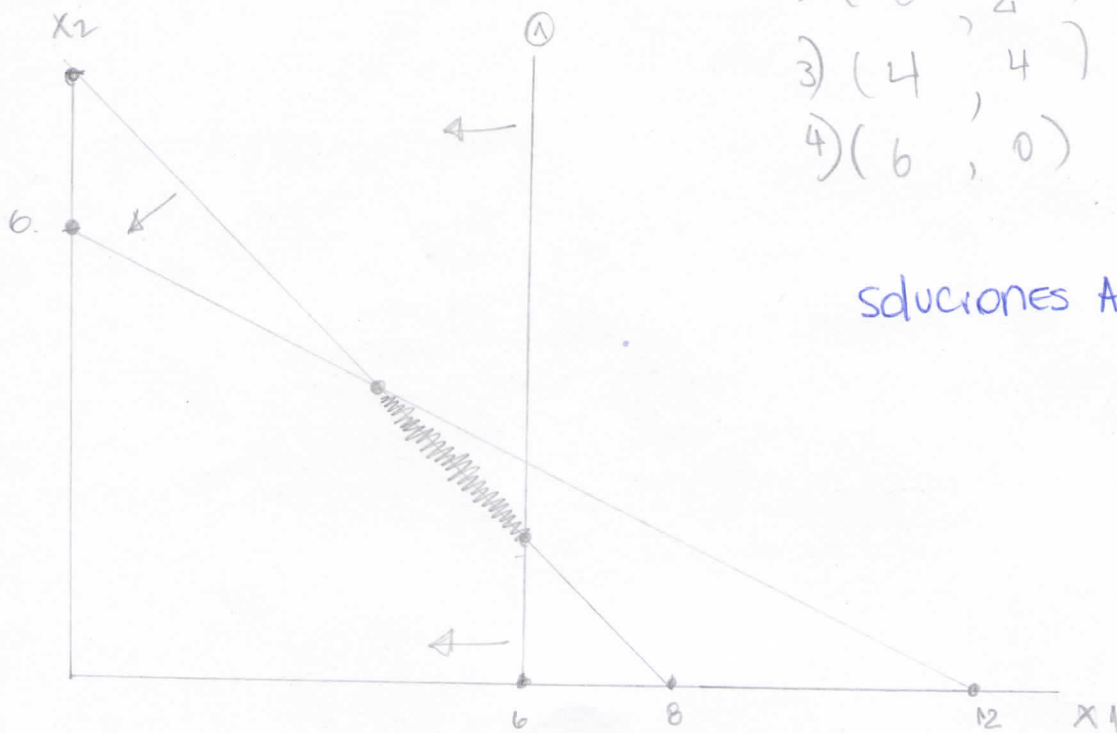
$$Z = 3 X1 + X2 \quad \text{MAX}$$

infeasible: El modelo es incompatible (no tiene solución válida)
unbounded: El modelo es un polígono abierto.

$$\begin{cases} x_1 = 6 & \textcircled{1} \\ x_1 = 8 \text{ y } x_2 = 8 & \textcircled{2} \\ x_1 = 12 \text{ y } x_2 = 6 & \textcircled{3} \end{cases} \quad z = 4x_1 + 4x_2 \text{ (Max)}$$

$$\begin{array}{l} x_1 \quad x_2 \\ 1) (0, 6) \\ 2) (6, 2) \\ 3) (4, 4) \\ 4) (6, 0) \end{array}$$

Grafico 1.7



soluciones Alternativas

$$\begin{aligned} x_1 + x_2 &= 8 \\ 6 + x_2 &= 8 \\ x_2 &= 8 - 6 = 2 \end{aligned}$$

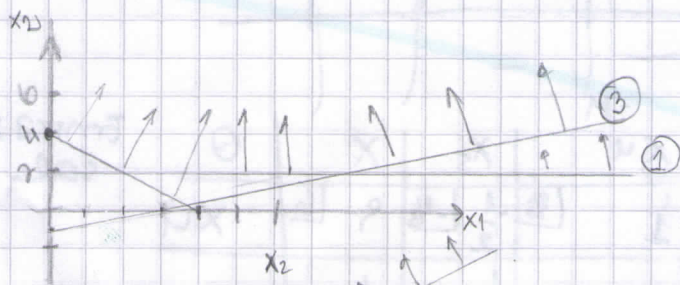
$$\begin{aligned} x_1 &= 8 - x_2 \rightarrow x_1 = 8 - 4 = 4 \\ x_1 + 2x_2 &= 12 \\ 8 - x_2 + 2x_2 &= 12 \\ x_2 &= 12 - 8 \\ x_2 &= 4 \end{aligned}$$

$$\begin{aligned} 1) \quad z &= 4 \cdot 0 + 4 \cdot 6 \\ z &= 24 \end{aligned}$$

$$\begin{aligned} 2) \quad z &= 6 \cdot 4 + 2 \cdot 4 \\ z &= 32 \end{aligned}$$

$$3) \quad z = 4 \cdot 4 + 4 \cdot 4 \Rightarrow z = 32$$

$$4) \quad 6 \cdot 4 = z = 24$$



1.10

→ $x_2 = 2$ ①

→ $x_1 = 6, x_2 = 4$ ②

→ $x_1 = 3, x_2 = -1$ ③

Es un poligono abierto.

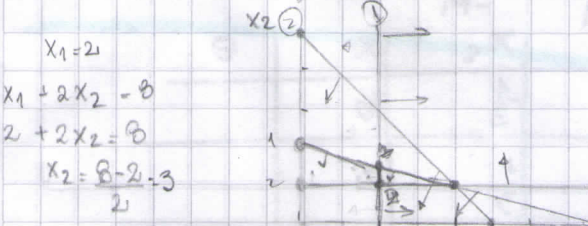
solucion incompatible

1.12

① $x_1 = 6, x_2 = 6$

② $x_1 = 1/2, x_2 = 1/3$

③ $x_1 = -8, x_2 = 4$



1.11

$x_1 \geq 0$

$x_1 = 2$
 $x_1 + 2x_2 = 8$
 $2 + 2x_2 = 8$
 $x_2 = 8 - 2 = 3$
 $x_2 = 3$

$z(\text{Max}) = -x_1 + 2x_2$
 ① $-2 + 2 \cdot 3 = 4 \leftarrow \text{Max } x^*(2, 3) \quad z = 4$
 ② $-2 + 2 \cdot 2 = 2$
 ③ $-4 + 2 \cdot 2 = 0$

3 (4, 2) z =
 1 (2, 3) z =
 2 (2, 2) z =

$x_1 = 2$ ①
 $x_1 = 5, x_2 = 10$ ②
 $x_1 = 8, x_2 = 4$ ③
 $x_2 = 1$ ④

1.13

① $x_2 = 8, x_1 = 4$

② $x_1 = 6, x_2 = 6$

③ $x_1 = 4$

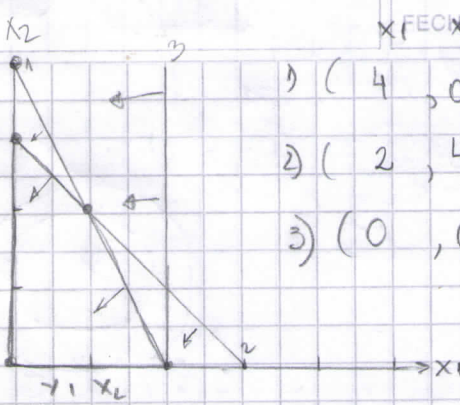
$z = 3x_1 + x_2$

① $3 \cdot 4 = 12$

$z = 12 \quad x^* = (4, 0)$

② $2 \cdot 3 + 4 = 10$

③ $6 = 6$



HOJA N°

FECHA

1) (4, 0) z =
 2) (2, 4) z =
 3) (0, 6) z =